



Geometric Sequences and Series

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Learning Objectives

- Identify geometric sequences and find the common ratio
- Apply the explicit formula for the n th term of a geometric sequence
- Find specified terms of a geometric sequence
- Compute finite and infinite geometric series sums

Solve each problem, showing all work and writing your final answer in simplest form.

1. Determine whether the sequence is geometric. If yes, state the common ratio.

3, 6, 12, 24, 48, ...

Answer: _____

2. Find the common ratio of the geometric sequence.

81, 27, 9, 3, ...

Answer: _____

3. Write the explicit formula for the n th term of the geometric sequence with first term 5 and common ratio 2.

$a_1 = 5, r = 2$

Answer: _____

4. Find the 8th term of the geometric sequence.

$a_1 = 2, r = 3$

Answer: _____

5. Find the 6th term of the geometric sequence.

4, -8, 16, -32, ...

Answer: _____

6. Find the first term of a geometric sequence given the common ratio and the 5th term.

$r = 2, a_5 = 48$

Answer: _____

7. Find the sum of the first 7 terms of the geometric sequence.

$a_1 = 3, r = 2$

Answer: _____



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8. Find the sum of the first 5 terms of the geometric sequence.

6, 12, 24, 48, ...

Answer: _____

9. Evaluate the infinite geometric series.

$$\sum_{n=1}^{\infty} 8\left(\frac{1}{2}\right)^{n-1}$$

Answer: _____

10. Find the sum of the infinite geometric series, if it exists.

$$27 + 9 + 3 + 1 + \frac{1}{3} + \dots$$

Answer: _____





This worksheet covers Part 2 of sequences and series: identifying geometric sequences, finding the common ratio r , using the explicit formula $a_n = a_1 r^{(n-1)}$, finding specific terms, computing finite geometric series sums with $S_n = a_1(1 - r^n)/(1 - r)$, and evaluating infinite geometric series sums $S = a_1/(1 - r)$ when $|r| < 1$.

Solutions

1. Determine whether the sequence is geometric. If yes, state the common ratio.

3, 6, 12, 24, 48, ...

- Divide each term by the previous term to test for a constant ratio.
- Six divided by three equals two, twelve divided by six equals two, twenty-four divided by twelve equals two.
- Because the ratio is constant, the sequence is geometric with common ratio two.

Answer: Geometric, $r = 2$

2. Find the common ratio of the geometric sequence.

81, 27, 9, 3, ...

- Divide the second term by the first term.
- Twenty-seven divided by eighty-one equals one third.
- The common ratio is one third.

Answer: $r = \frac{1}{3}$

3. Write the explicit formula for the n th term of the geometric sequence with first term 5 and common ratio 2.

$a_1 = 5$, $r = 2$

- Recall the explicit formula for a geometric sequence: a sub n equals a sub one times r raised to the n minus one.
- Substitute a sub one equals five and r equals two.
- The formula is a sub n equals five times two to the n minus one.

Answer: $a_n = 5 \cdot 2^{n-1}$

4. Find the 8th term of the geometric sequence.

$a_1 = 2$, $r = 3$

- Use the formula a sub n equals a sub one times r raised to the n minus one.
- Substitute a sub one equals two, r equals three, and n equals eight.
- Two times three to the seventh power equals two times two thousand one hundred eighty-seven, which is four thousand three hundred seventy-four.

Answer: $a_8 = 4374$



5. Find the 6th term of the geometric sequence.

4, -8, 16, -32, ...

→ Find the common ratio by dividing negative eight by four to get negative two.

→ Apply a sub n equals a sub one times r raised to the n minus one with a sub one equals four, r equals negative two, and n equals six.

→ Four times negative two to the fifth power equals four times negative thirty-two, which is negative one hundred twenty-eight.

Answer: $a_6 = -128$

6. Find the first term of a geometric sequence given the common ratio and the 5th term.

$r = 2$, $a_5 = 48$

→ Use a sub five equals a sub one times r to the fourth power.

→ Substitute forty-eight equals a sub one times two to the fourth, which is sixteen.

→ Divide both sides by sixteen to get a sub one equals three.

Answer: $a_1 = 3$

7. Find the sum of the first 7 terms of the geometric sequence.

$a_1 = 3$, $r = 2$

→ Use the finite geometric series formula S sub n equals a sub one times the quantity one minus r to the n , divided by one minus r .

→ Substitute a sub one equals three, r equals two, and n equals seven: three times the quantity one minus one hundred twenty-eight, divided by one minus two.

→ Simplify the numerator to three times negative one hundred twenty-seven and divide by negative one to get three hundred eighty-one.

Answer: $S_7 = 381$

8. Find the sum of the first 5 terms of the geometric sequence.

6, 12, 24, 48, ...

→ Identify a sub one equals six and r equals two.

→ Apply the finite geometric series formula with n equals five.

→ Six times the quantity one minus thirty-two, divided by one minus two, gives six times thirty-one, which equals one hundred eighty-six.

Answer: $S_5 = 186$

9. Evaluate the infinite geometric series.

$$\sum_{n=1}^{\infty} 8\left(\frac{1}{2}\right)^{n-1}$$

→ Identify the first term a sub one equals eight and the common ratio r equals one half.

→ Because the absolute value of r is less than one, the series converges; use S equals a sub one divided by one minus r .

→ Eight divided by one minus one half equals eight divided by one half, which equals sixteen.

Answer: $S = 16$



10. Find the sum of the infinite geometric series, if it exists.

$$27 + 9 + 3 + 1 + \frac{1}{3} + \dots$$

→ The first term is twenty-seven and the common ratio is one third.

→ Since the absolute value of the common ratio is less than one, the sum converges.

→ Apply S equals a sub one divided by one minus r : twenty-seven divided by two thirds equals eighty-one over two.

Answer: $S = \frac{81}{2}$

